

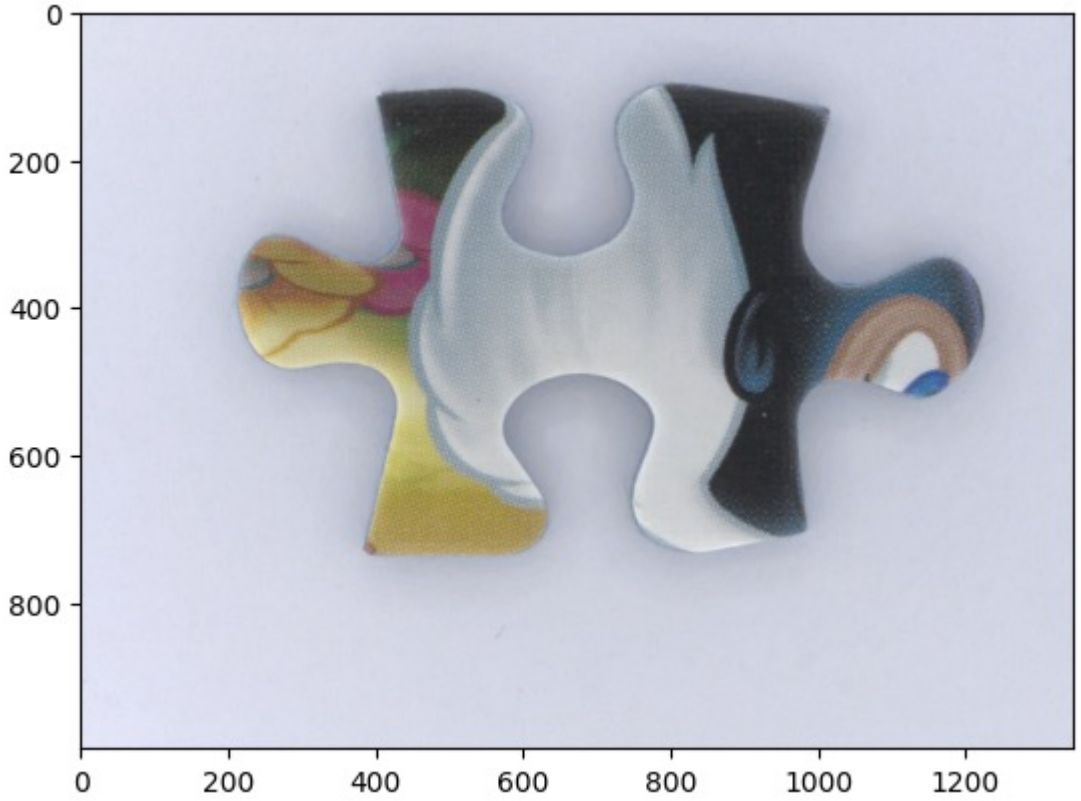
EDGE-DETECTION SEGMENTIERUNG UND ROS-NODE

ROBOTIK PROJEKTSTUDIUM
PROF DR. FISCHER - SS24

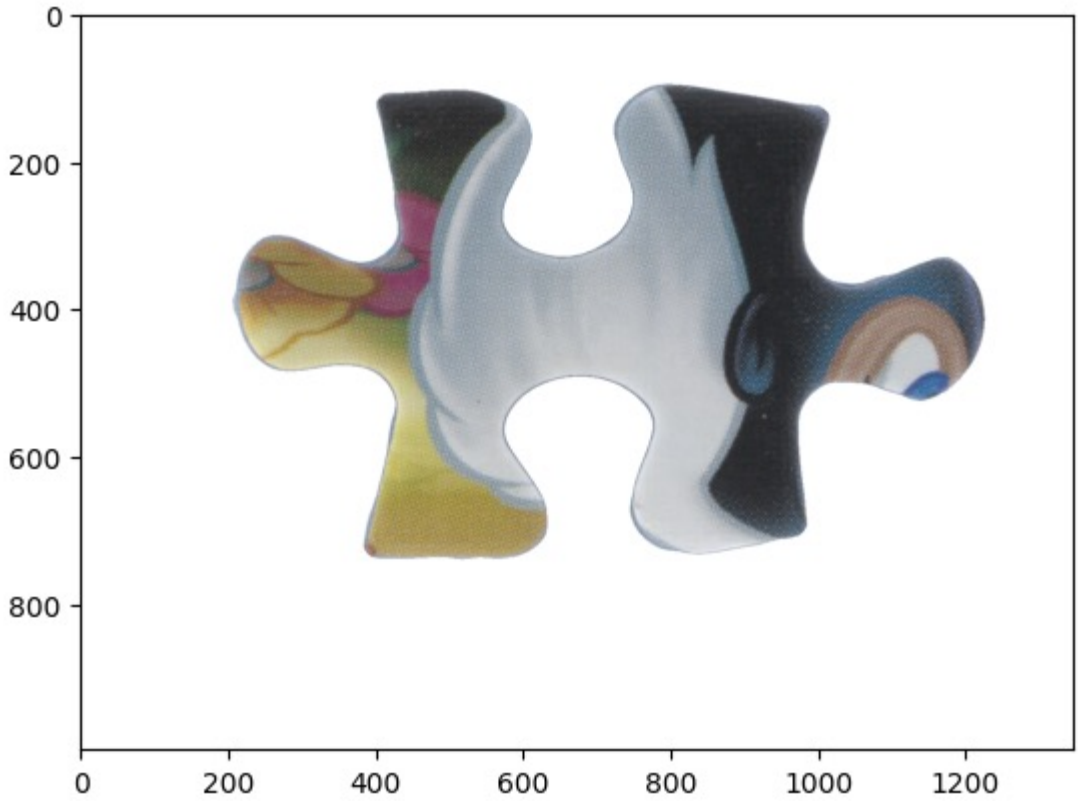
GLIEDERUNG

- Zielsetzung
- Verschiedene Verfahren
- Edge-Detection
  Segmentierung
- ROS-Node

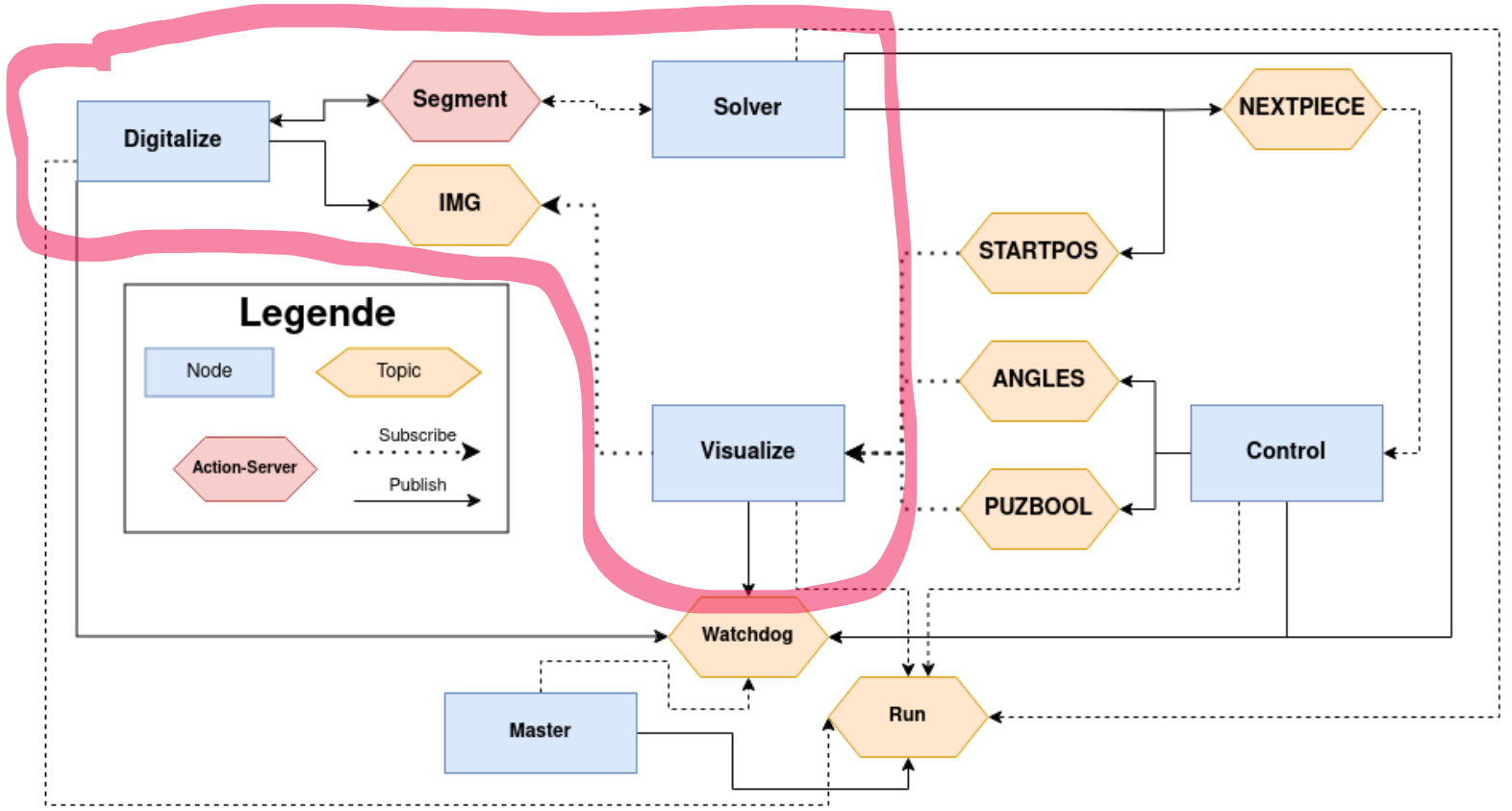
ZIELSETZUNG – SEGMENTIERUNG







ZIELSETZUNG - VERTEILUNG



VERSCHIEDENE VERFAHREN

Threshold Based – X

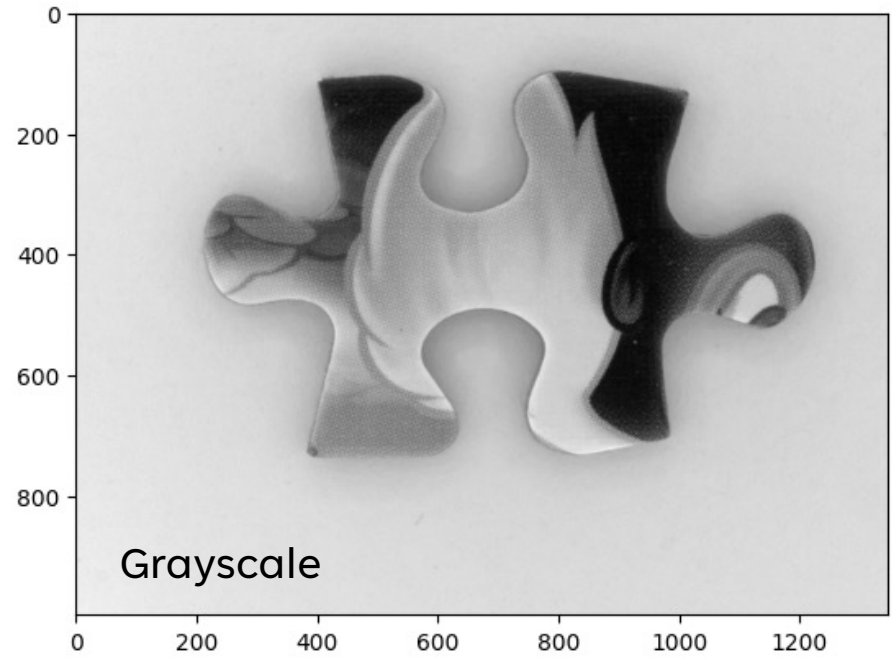
Watershed – X

Edge Detection - ✓

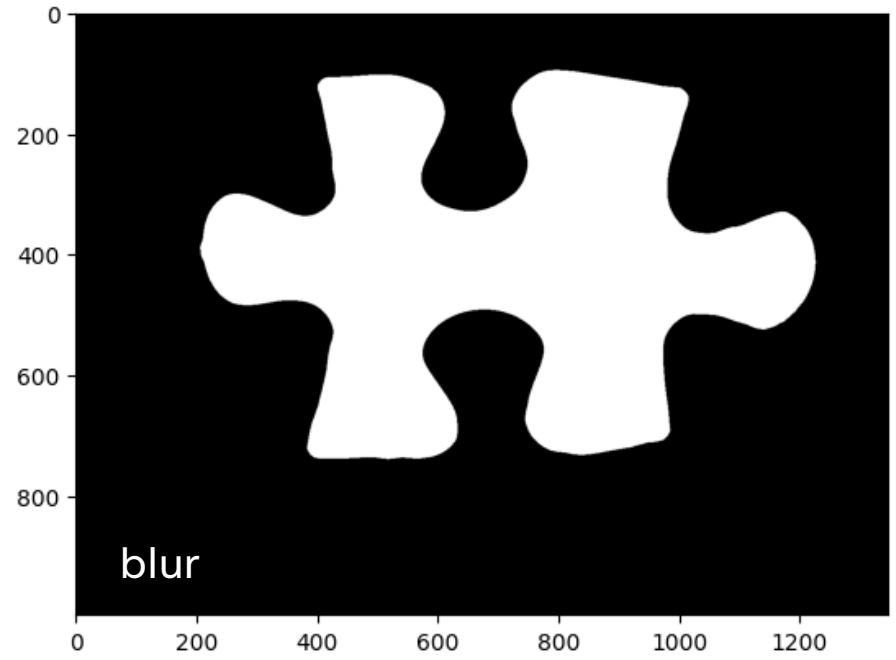
Aber, würde das Projekt noch weiter laufen…

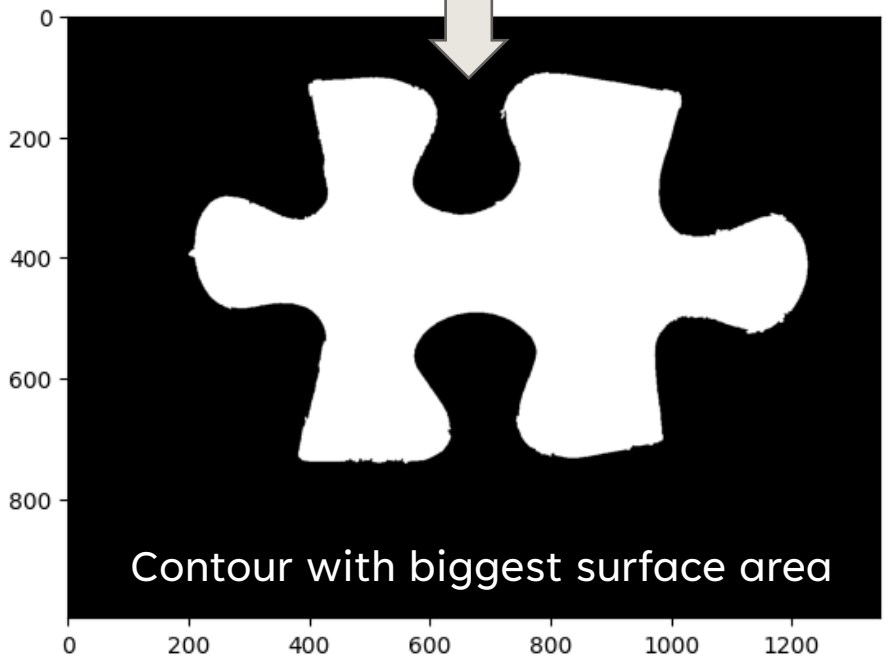
… watershed nochmal probieren!

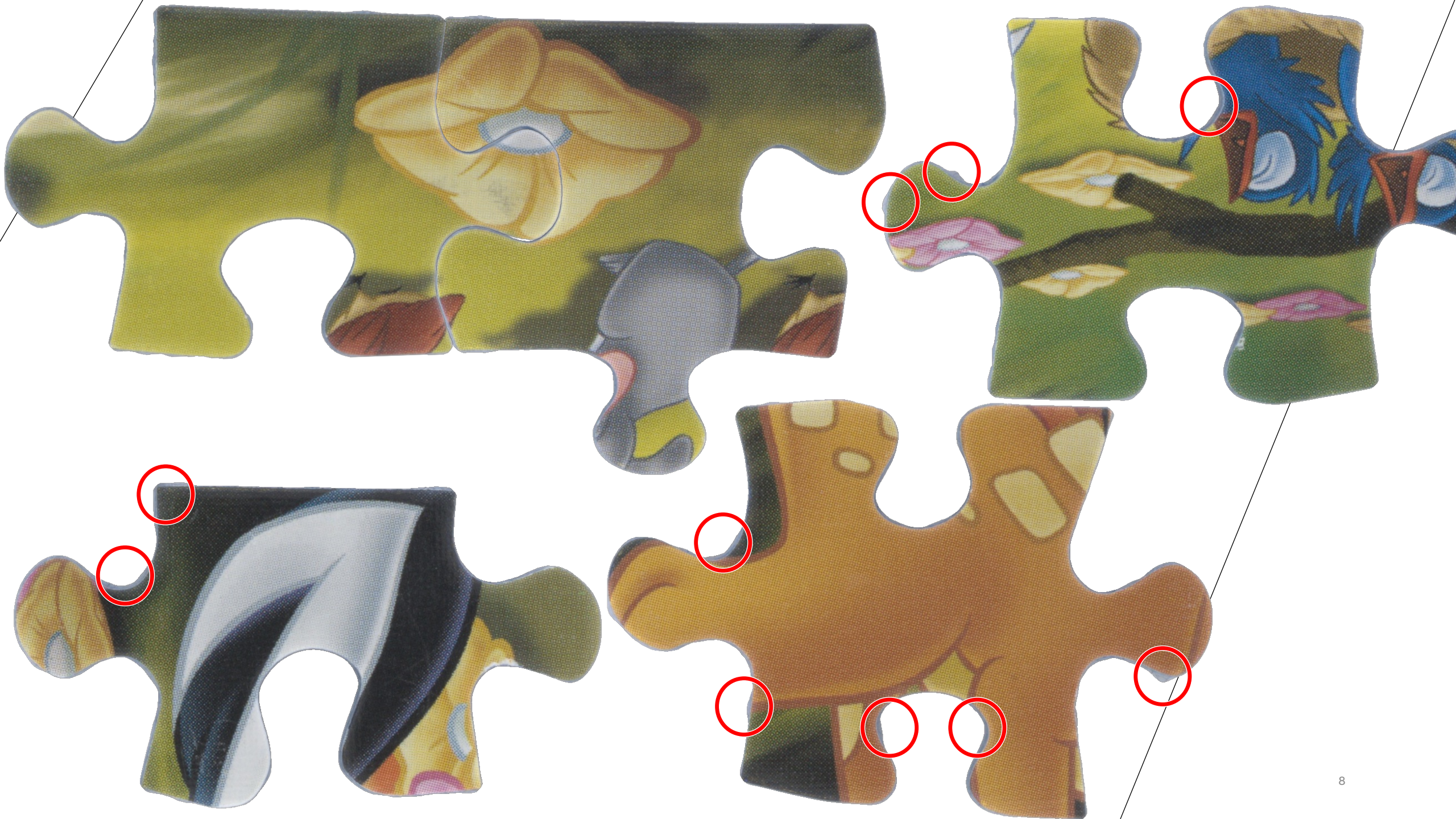
EDGE-DETECTION SEGMENTIERUNG


Grayscale


Enhanced edges with Canny


blur

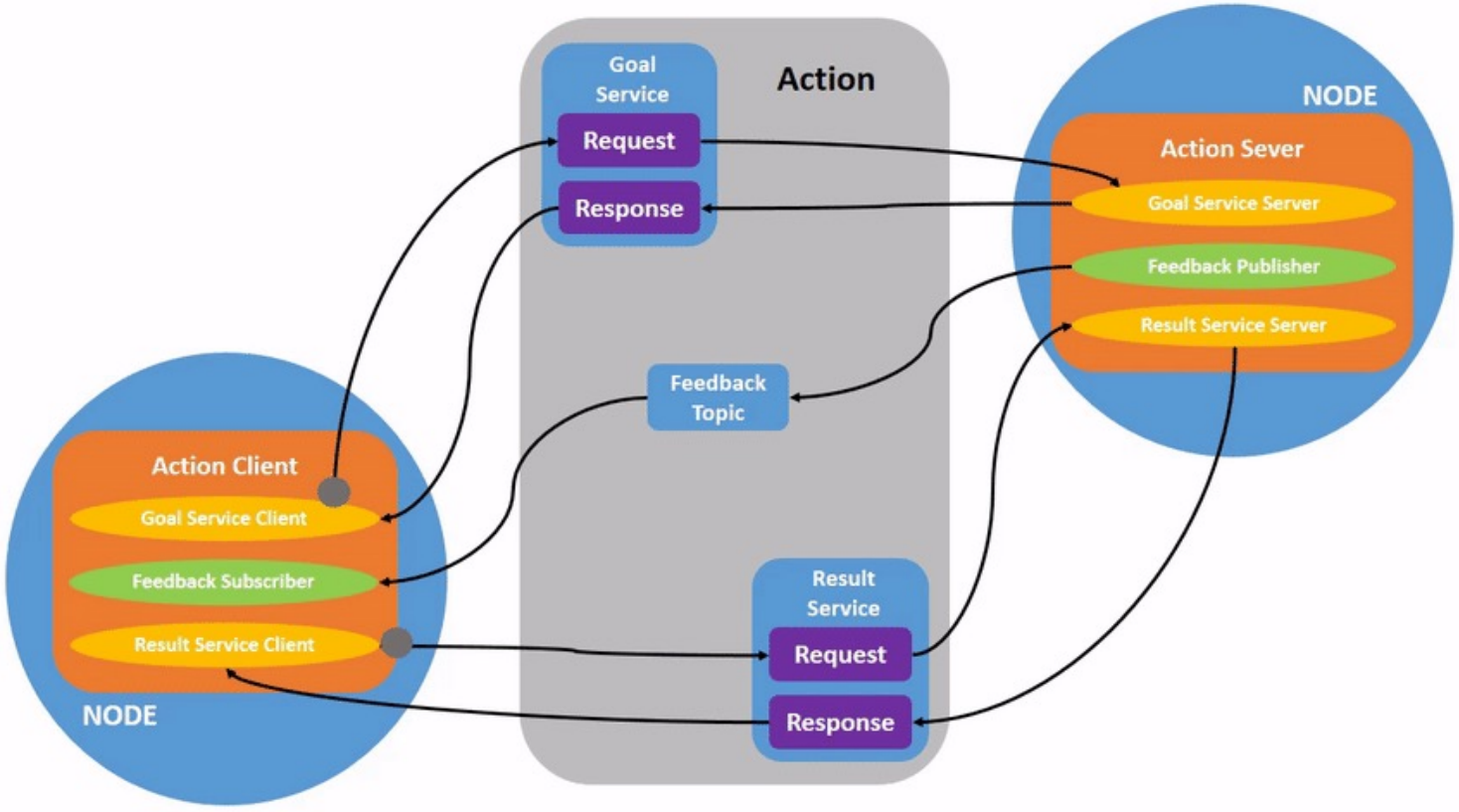

Contour with biggest surface area

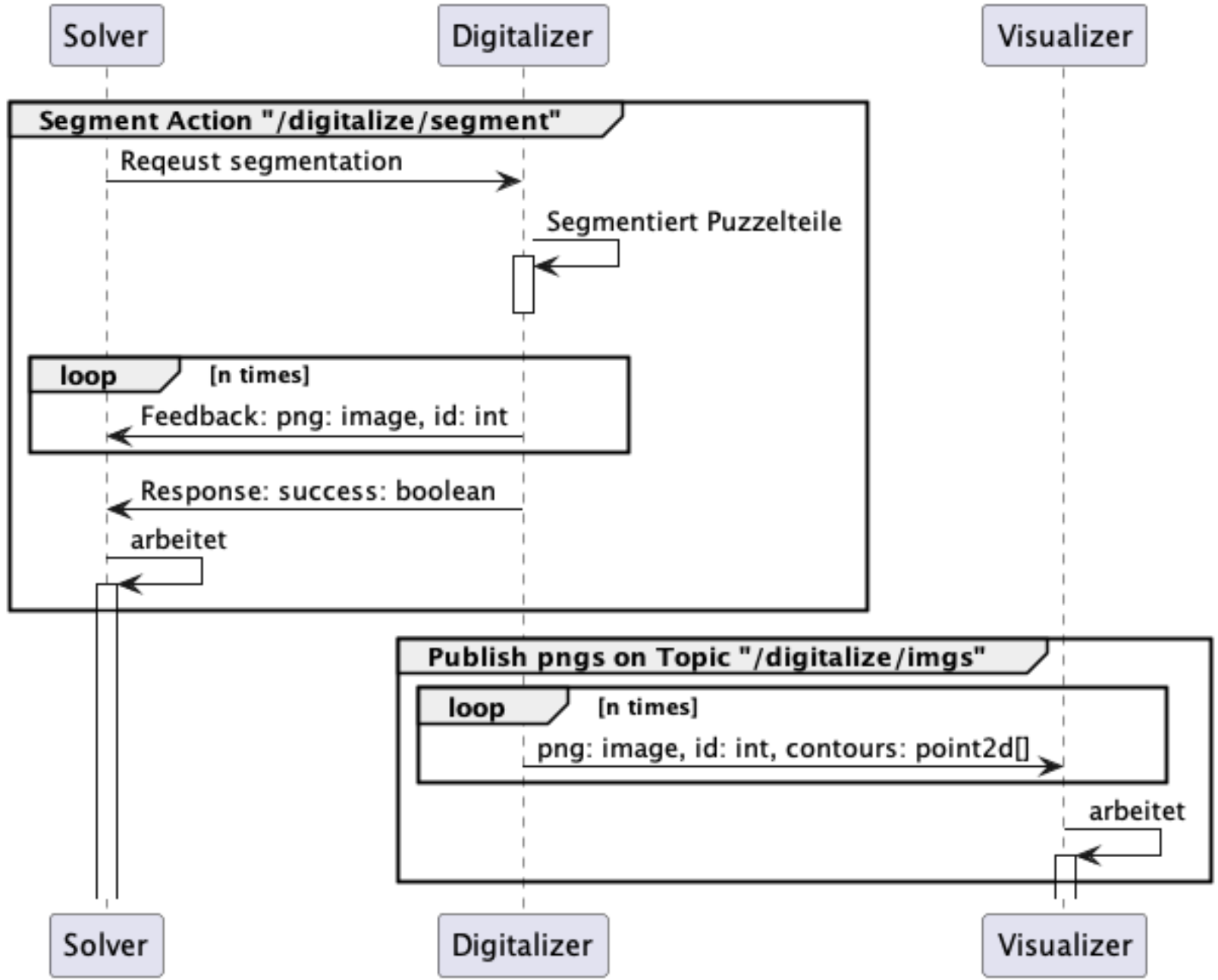




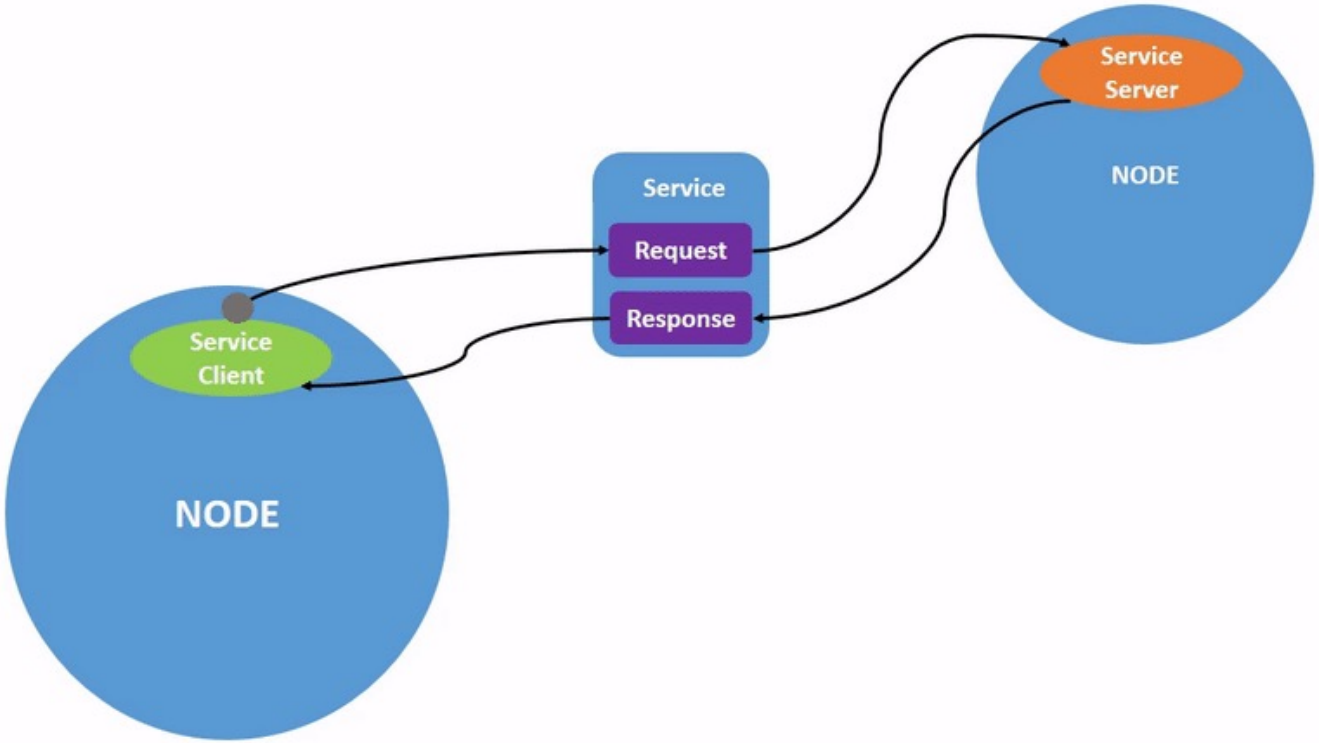
ROS-NODE

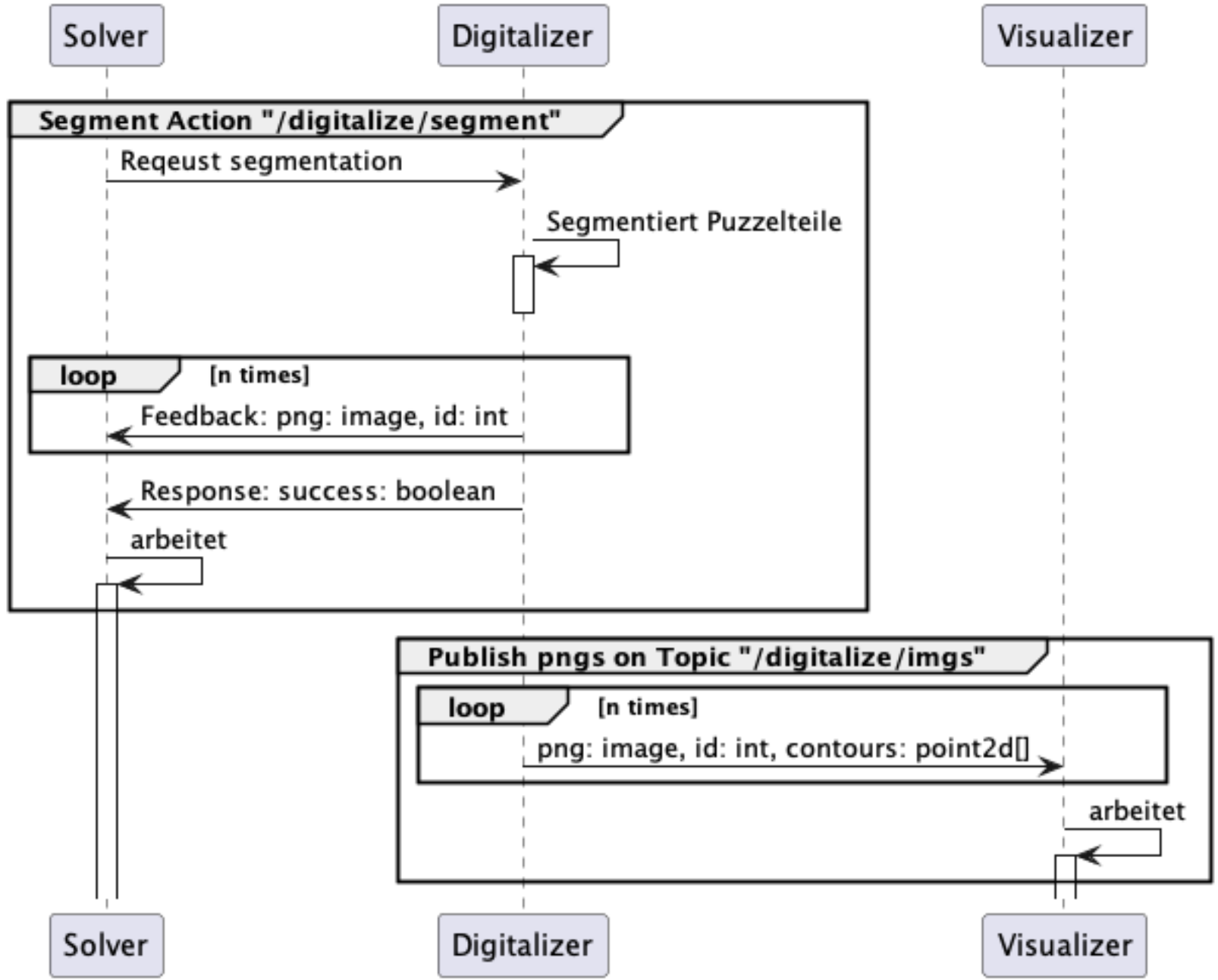
ROS NODE – KOMMUNIKATION – /DIGITALIZE/SEGMENT



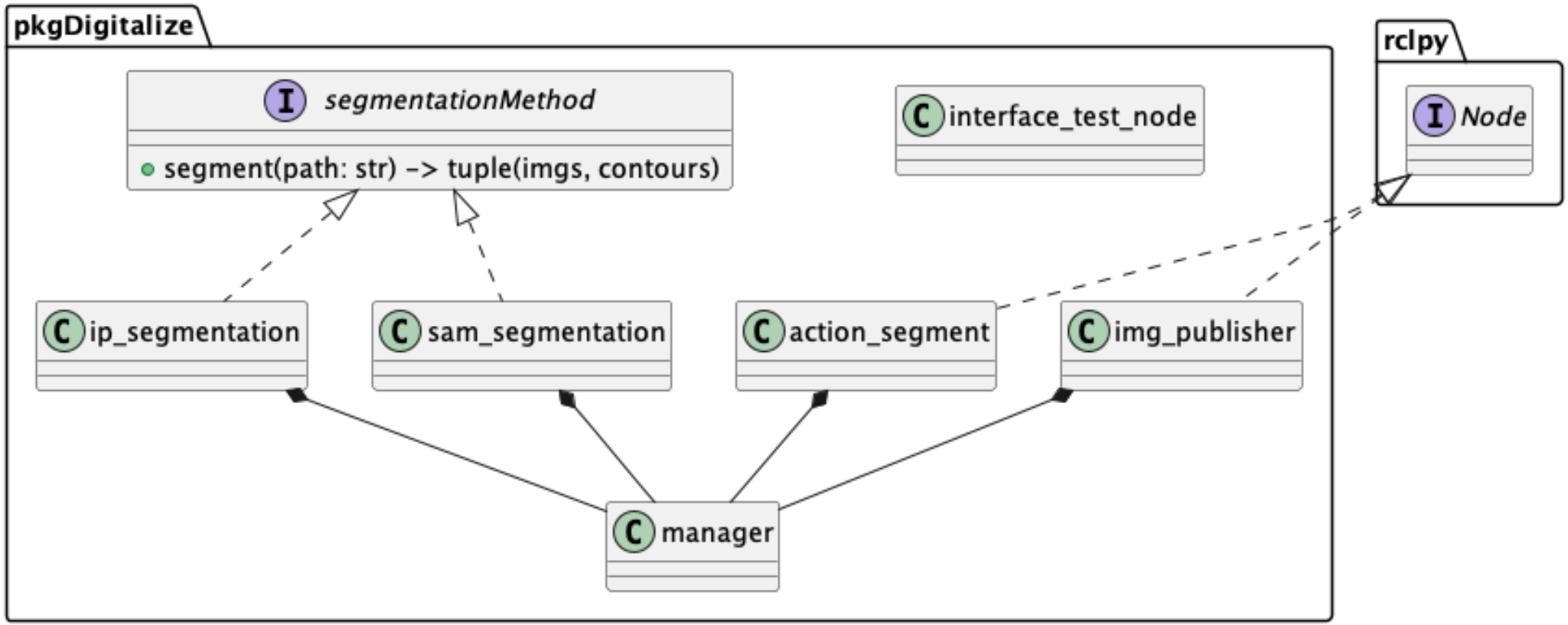

Solver
Digitalizer
Visualizer
Segment Action "/digitalize/segment"
Reqeust segmentation
Segmentiert Puzzelteile
loop
[n times]
Feedback: png: image, id: int
Response: success: boolean
arbeitet
Publish pngs on Topic "/digitalize/imgs"
loop
[n times]
png: image, id: int, contours: point2d[]
arbeitet
Solver
Digitalizer
Visualizer

ROS NODE – KOMMUNIKATION - /DIGITALIZE/IMGS




Solver
Digitalizer
Visualizer
Segment Action "/digitalize/segment"
Reqeust segmentation
Segmentiert Puzzelteile
loop
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arbeitet
Solver
Digitalizer
Visualizer

ROS NODE - ARCHITEKTUR



VIELEN DANK FÜR IHRE AUFMERKSAMKEIT!

Korbinian Kipp